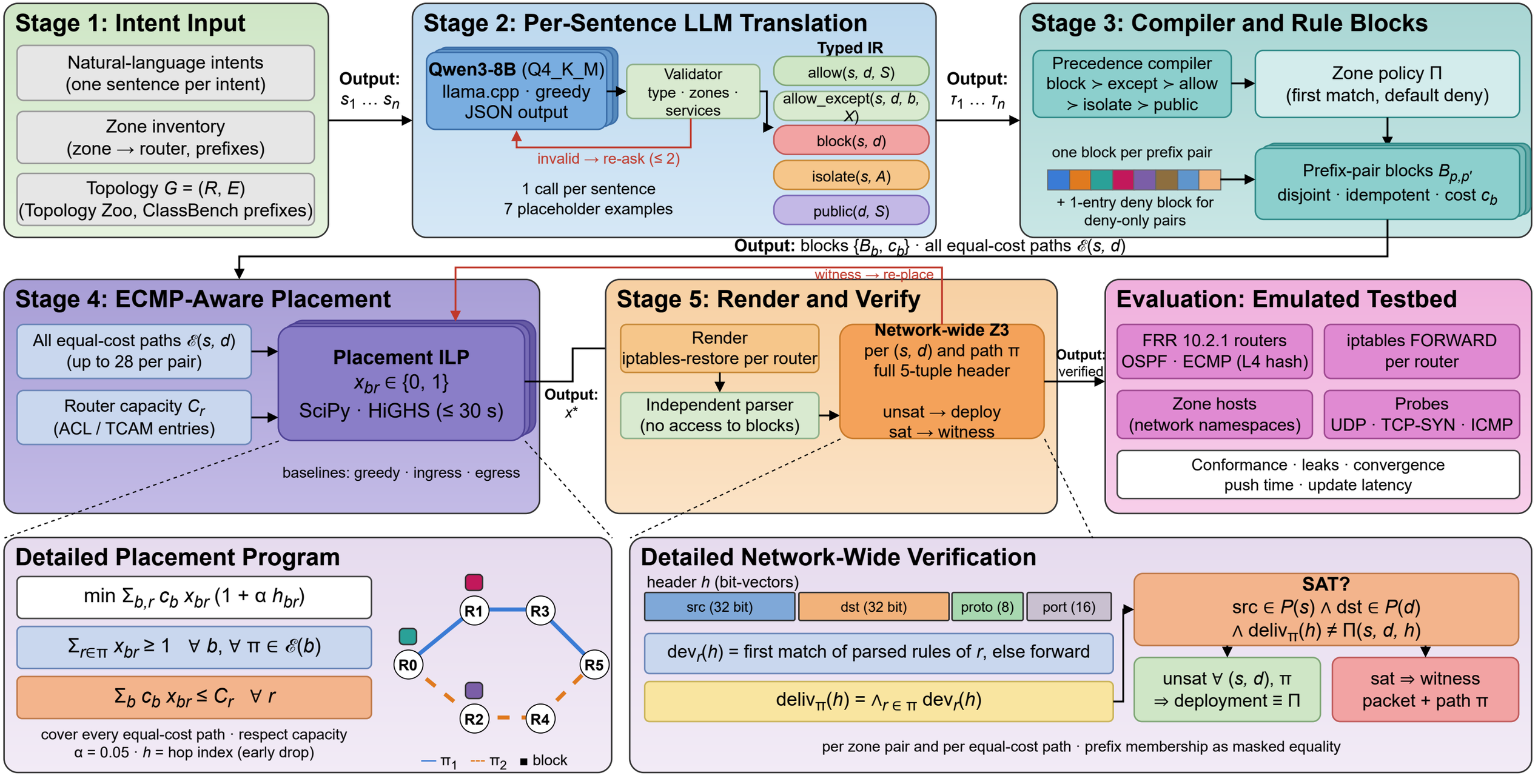

Stage 1: Intent Input
Natural-language intents
(one sentence per intent)
Zone inventory
(zone → router, prefixes)
Topology G = (R, E)
(Topology Zoo, ClassBench prefixes)
Output:
s1 … sn
Stage 2: Per-Sentence LLM Translation
Qwen3-8B (Q4_K_M)
llama.cpp · greedy
JSON output
Validator
type · zones · services
invalid → re-ask (≤ 2)
1 call per sentence
7 placeholder examples
Typed IR
allow(s, d, S)
allow_except(s, d, b, X)
block(s, d)
isolate(s, A)
public(d, S)
Output:
τ1 … τn
Stage 3: Compiler and Rule Blocks
Precedence compiler
block ≻ except ≻ allow
≻ isolate ≻ public
Zone policy Π
(first match, default deny)
one block per prefix pair
+ 1-entry deny block for
deny-only pairs
Prefix-pair blocks Bp,p'
disjoint · idempotent · cost cb
Output: blocks {Bb, cb} · all equal-cost paths 𝒫(s, d)
witness → re-place
Stage 4: ECMP-Aware Placement
All equal-cost paths 𝒫(s, d)
(up to 28 per pair)
Router capacity Cr
(ACL / TCAM entries)
Placement ILP
xbr ∈ {0, 1}
SciPy · HiGHS (≤ 30 s)
baselines: greedy · ingress · egress
Output:
x*
Stage 5: Render and Verify
Render
iptables-restore per router
Independent parser
(no access to blocks)
Network-wide Z3
per (s, d) and path π
full 5-tuple header
unsat → deploy
sat → witness
Output:
verified
Evaluation: Emulated Testbed
FRR 10.2.1 routers
OSPF · ECMP (L4 hash)
iptables FORWARD
per router
Zone hosts
(network namespaces)
Probes
UDP · TCP-SYN · ICMP
Conformance · leaks · convergence
push time · update latency
Detailed Placement Program
min Σb,r cb xbr (1 + α hbr)
Σr∈π xbr ≥ 1 ∀ b, ∀ π ∈ 𝒫(b)
Σb cb xbr ≤ Cr ∀ r
cover every equal-cost path · respect capacity
α = 0.05 · h = hop index (early drop)
R0
R1
R2
R3
R4
R5
π1
π2
block
Detailed Network-Wide Verification
header h (bit-vectors)
src (32 bit)
dst (32 bit)
proto (8)
port (16)
devr(h) = first match of parsed rules of r, else forward
delivπ(h) = ∧r ∈ π devr(h)
SAT?
src ∈ P(s) ∧ dst ∈ P(d)
∧ delivπ(h) ≠ Π(s, d, h)
unsat ∀ (s, d), π
⇒ deployment ≡ Π
sat ⇒ witness
packet + path π
per zone pair and per equal-cost path · prefix membership as masked equality